

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

December 2013

EC308 Antenna and Wave Propagation

Date: 3.12.2013, Tuesday Time: 1.30 p.m. To 4.30p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

- Q-1 Attempt the following terms (Any seven). 07
- (i) Directivity
 - (ii) Beam Solid Angle
 - (iii) HPBW
 - (iv) End fire array
 - (v) Beam width of an antenna
 - (vi) Antenna beam efficiency
 - (vii) Gain
 - (viii) Radiation resistance
- Q-2 Attempt the following questions. 14
- (i) Derive an expression for the resultant field of an array of N isotropic sources of equal amplitude and spacing and obtain the condition for operating it as Broad side array.
 - (ii) Explain the concept of principle of pattern multiplication with an example.
- OR**
- Q-2 Attempt the following questions. 14
- (i) Derive an expression for the resultant field of an two isotropic sources of same amplitude and phase and draw field pattern for the same.
 - (ii) Write a short notes on (a) Binomial arrays (b) Phased array
- Q-3 Attempt the following questions. 14
- (i) Derive expression for Radiation Resistance of Loop antenna.
 - (ii) Explain Normal & Axial Mode of Helical antenna.
- OR**
- Q-3 Attempt the following questions. 14
- (i) Derive the expression for radiation resistance of a half-wave dipole Antenna.
 - (ii) State and Explain Dolph - Tchebysheff distribution for linear arrays.

SECTION II

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| Q-4 (a) | Define the following terms.
(i) Virtual height
(ii) Maximum Usable Frequency,
(iii) Skip Distance | 03 |
| (b) | Figure out Different ionized regions | 04 |
| Q-5 (a) | List out the advantages and disadvantages of Micro strip Patch antennas | 07 |
| (b) | Explain the Yagi Uda antenna. | 07 |
| OR | | |
| Q-5(a) | Discuss Babinet's Principle. | 07 |
| (b) | Explain reflector Lens antenna | 07 |
| | | |
| Q-6(a) | Discuss the different feeding methods to reflector antenna | 07 |
| (b) | How horizontal antenna and Vertical antenna work above the ground. | 07 |
| OR | | |
| Q-6(a) | Explain the types of antenna measurements. | 07 |
| (b) | Attempt the following examples. | 07 |
| (i) | Calculate the maximum aperture of an antenna which is operating at a wavelength of 2 m and $D=100$. | |
| (ii) | Determine the maximum effective aperture of a beam antenna having HPBW of 30° and 35° in perpendicular planes intersecting in the beam axis. | |